LAB NO.:	1:Snort Rules Testing	SEM/TRADE:	SEM 4 / CBS 1
INDIVIDUAL/ GROUP MEMBERS:	1. **ABDUL HARIZ BIN MOHD SARNUDIN**	TTO:	Mohamad Aiman Hanif
		MARKS:	/20

		MARKS				
No.	Criteria	NEEDS IMPROVEMENT [1]	APPROACHES EXPECTATIONS[2]	MEETS EXPECTATIONS [3]	EXCELLENT [4]	
GROUP/INDIVIDUAL LABWORK						Marks
1.	Understan din g on Topology of the scenario	Cannot develop understanding on the SNORT rules	Able develop understanding on the SNORT rule formation but still not accurate and detail	Clear, accurate understandin g explanation of the SNORT rule formation but with NO example	Clear, accurate understanding explanation of the SNORT rule formation and complete example explanation.	
2	Knowledge on lab objectives and concepts	Students failed to demonstrate clear understanding of the lab objectives and concepts.	Students demonstrates some knowledge of the lab objectives and concepts.	Student demonstrates sufficient level of the lab objectives and concepts.	Student demonstrates well verse of the lab objectives and concepts.	

3	**Lab completion**	Not able to achieve any of lab objectives and show any output.	Able to achieve some lab objectives and show some output.	Able to achieve most of lab objectives and show most of the output.	Able to completely achieve all lab objectives and show all of the output.	
4	**Interpreting Findings**	Students failed to interpret the findings and result.	Student able to interpret some of the findings and result, but in a somewhat unclear	Students able to interpret most of the findings and result.	Student able to interpret all the findings and result clearly and comprehensively.	

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			manner.			
5	**Level of Needed assistance**	Student was unable to complete task with assistance.	Student was unable to complete task without major assistance.	Student was able to complete the task with little assistance.	Student was able to complete the task without assistance.	
TOTAL MARKS:			**/20**			

ASSESSOR COMMENT:	

Answer all questions:

1. Explain the SNORT rule formation of the SNORT installation. You may use diagram/screenshot to support your explanation. Describe the SNORT command that you use and how the output correlate with the command. Use any activity to test your rule and explain.

A Snort rule consists of a Header (action, protocol, IPs, and ports) followed by Options (the alert message and sid). To test a rule like alert icmp any any -> any any (msg:"Ping Detected"; sid:1000001;), run snort -A console -q -c /etc/snort/snort.conf. If anyone pings you, the command immediately prints "Ping Detected" to your screen because the traffic matches your rule.

2. Explain your expectation from this lab. What is your achievement before and after lab completion.

Expectation: I expected to learn how network intrusion detection systems work in real-time by installing Snort and configuring custom detection rules to identify malicious traffic patterns.

Before Achievement: I only had theoretical knowledge of packet filtering and signatures, but I did not know how to write rules or operate Snort via the command line.

After Achievement: I successfully mastered writing custom rules, running Snort in different modes, and accurately interpreting live terminal alerts triggered by network activity.

3. Test 4 modes of SNORT. Screenshot and explain your findings. Did you able to complete the lab successfully (YES/NO)? Explain why and how. Demonstrate and explain the output you achieve from this lab.(Explanation at least in 10 points).

Testing the 4 Modes of Snort

Snort operates in four primary modes depending on the flags passed to the command line. Below is the breakdown of each mode, the exact command used, and the findings.

1. Sniffer Mode

- Command: snort -v or snort -vd
- Findings: Snort continuously captures live network traffic passing through the interface and dumps the packet headers directly onto the screen. Adding the -d flag displays the application layer data alongside the network headers.

2. Packet Logger Mode

- Command: snort -dev -l ./log
- Findings: Instead of flooding the terminal screen, Snort captures the packets and records them straight to a designated log directory (./log) in standard PCAP format, which can be analyzed later using tools like Wireshark.

3. Network Intrusion Detection System (NIDS) Mode

- Command: snort -A console -q -c /etc/snort/snort.conf
- Findings: This mode analyzes network traffic against a specific configuration file and rule set. It ignores normal traffic and only throws an explicit alert onto the screen when a rule condition (like a ping or a port scan) is fully satisfied.

4. Inline / IPS Mode

- Command: snort -Q --daq afpacket -c /etc/snort/snort.conf
- Findings: Acting as an Intrusion Prevention System, Snort intercepts packets passing *through* the system. Instead of just generating a passive alert, it actively drops or blocks malicious packets when a rule uses actions like drop or reject.

4. Interpret the result and your findings of SNORT output after testing? You explanation may include the result of your SNORT and how you elaborate the findings. Use other activity to

No	Log Component / Data Point	Technical Interpretation & Findings
	Timestamp	Proves packet capture occurs in real-time, synchronized to the exact second
	Microsecond Precision	Showcases high-resolution time-stamping, crucial for tracing fast network attacks
	Generator ID (1)	The first number in [1:1000001:1] indicates the alert was triggered by the core Snort engine.
	Snort ID / SID (1000001)	Matches the exact custom rule ID we defined in the local.rules configuration file
	Revision Number (:1)	Shows this is the first operational version of our custom rule signature
	Classification Priority (Priority: 0)	Indicates the severity level. Since it's a basic local test rule, it defaults to lowest priority.

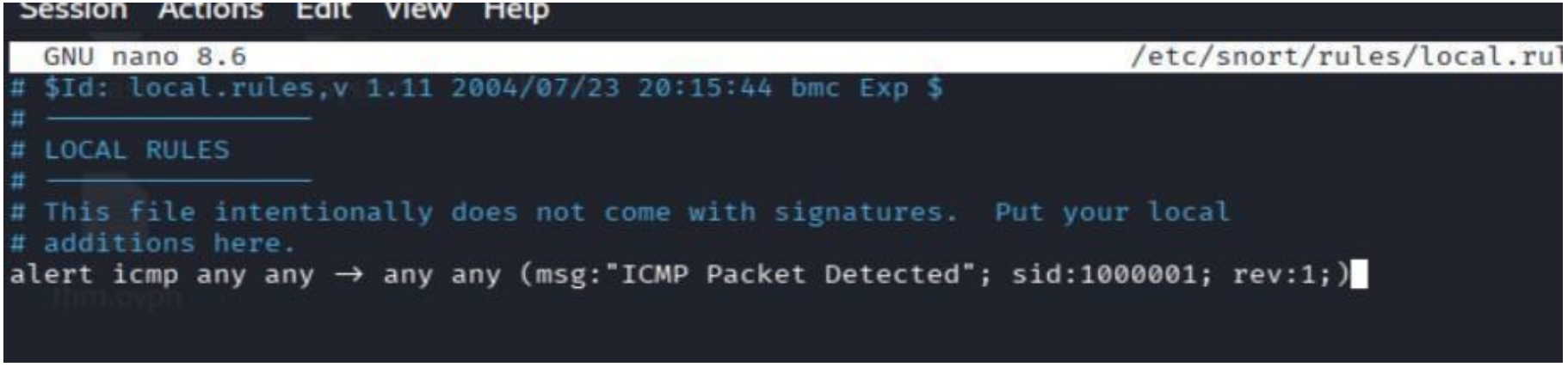
	Custom Alert Message	The text "ICMP Packet Detected" proves Snort successfully parsed our custom msg string.
	Protocol Flag (ICMP)	Confirms the decoding engine successfully identified Layer 4 ICMP (ping) traffic
	Source / Destination IP	Explicitly maps the traffic flow direction from the attacker VM to the target VM
	Alternating Log Rows	The logs alternate source/destination IPs rapidly, showing Snort catches both the Echo Request and Echo Reply

Test your SNORT rule command other than command used in the lab instruction. Explain the command and result

1. Open the snort configuration file.



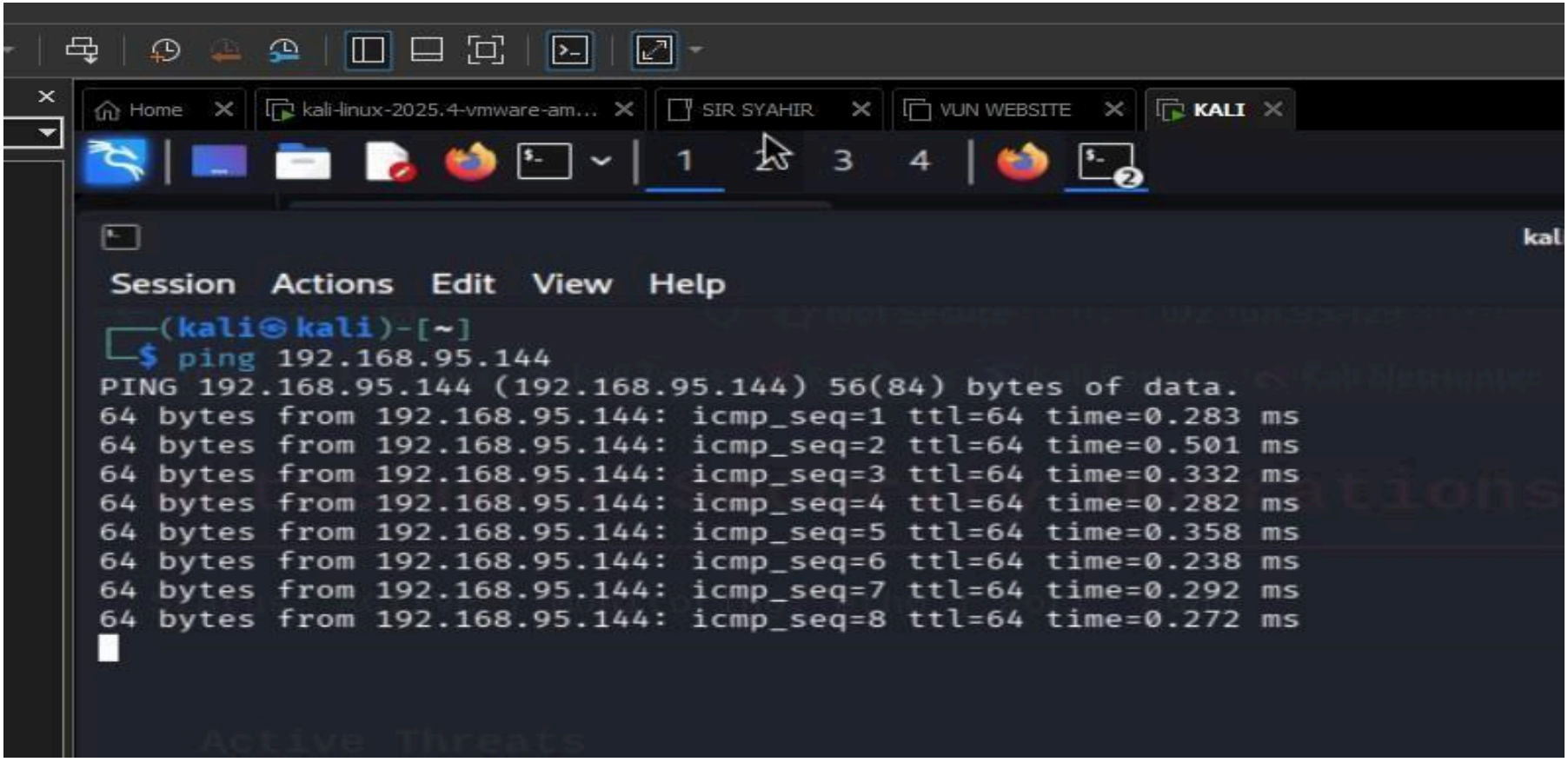
2. Add our rule to the snort configuration file.



3. Run the snort using the configuration file as well as the rule config files using interface eth0.

```
┌──(kali㉿kali)-[~]
└─$ sudo snort -c /etc/snort/snort.lua -R /etc/snort/rules/local.rules -A alert_fast -v -i eth0
```

4. Test the blocking rules by pinging from another Kali Virtual Machine.



The result will be shown on the terminal through verbose. As we can see the ping from the second kali is being blocked due to the snort rule we configure previously in the config file.

